



Nottingham Trent
University

Department of Computer Science

COMP40771 – Computational Intelligence

Lecture 7: Agentic AI - Multi-Agent Systems, Autonomy & Decision-Making

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Nottingham Trent
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Lecture Outline

In this lecture will cover:

- Define AI agents and Agentive AI
- Explain autonomy and rational decision-making
- Understand MAS foundations
- Analyse coordination and decision frameworks
- Evaluate safety, governance, and deployment issues

This lecture contributes to the following learning outcomes:

MLO2: Critically evaluate the strengths and weaknesses of different computational intelligence techniques for solving complex problems.

MLO4: Understand the ethical implications of using computational intelligence technologies and the potential impact on society.

MLO5: Apply computational intelligence techniques to solve real-world problems, demonstrating practical problem-solving skills.

MLO6: Develop proficiency in programming and using computational tools for implementing and evaluating intelligent systems.

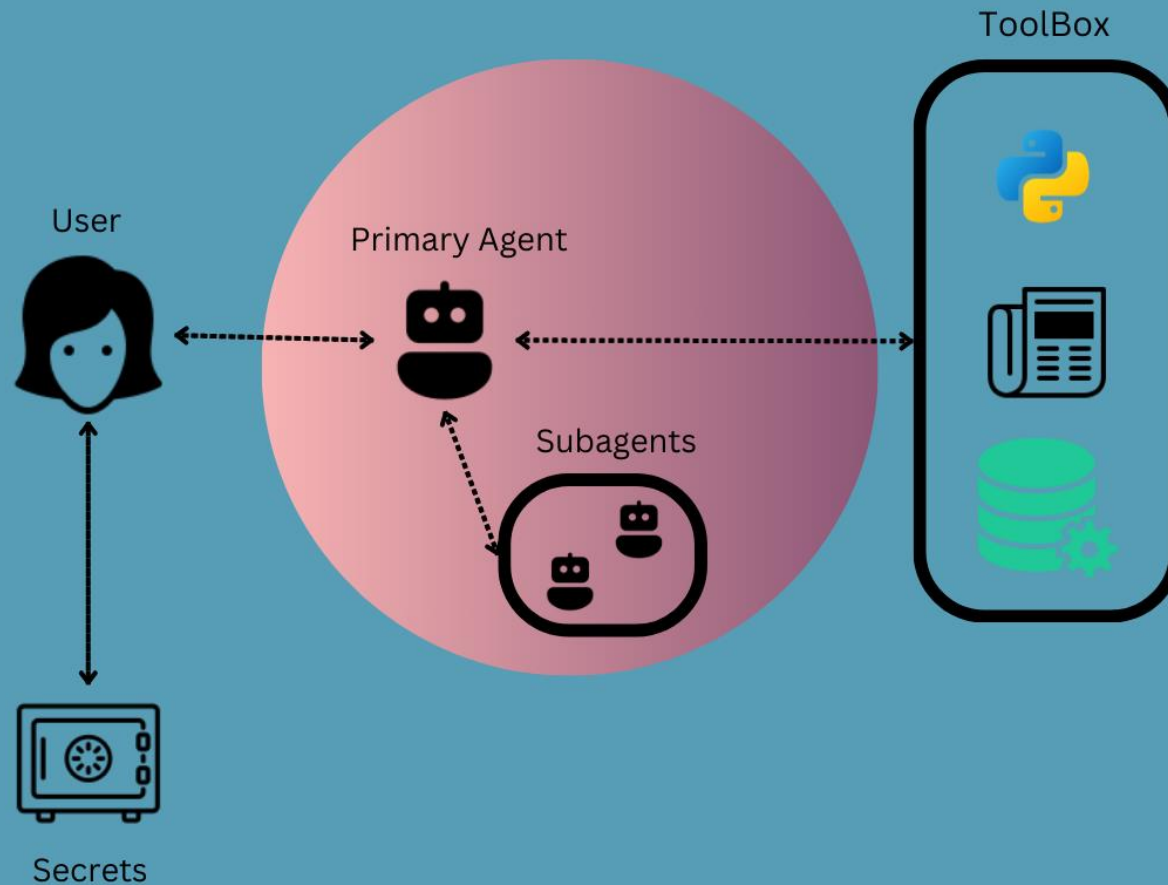
MLO8: Communicate effectively the concepts, methodologies, and results of computational intelligence research to both technical and non-technical audiences.

From Prediction to Action

- Traditional AI: focused on classification, regression, and prediction (e.g., image recognition, fraud detection).
 - Outputs are typically single-step decisions or scores, not extended action sequences.
- Generative AI: synthesises new content (text, code, images) using learned probabilistic models.
 - Generation improves reasoning capabilities but remains largely prompt-driven and reactive.
- Agentic AI: moves beyond output generation to goal pursuit through planning, task decomposition, tool use, and multi-step execution



AI Agent Architecture



AI Agents

- **Perception:** receives information from its environment through sensors, data streams, or inputs.
- **Internal state:** maintains a representation of the environment or context (memory, beliefs, world model).
- **Policy:** decision function that maps perceived states to actions (rule-based, optimisation-based, or learned).
- **Action:** executes decisions to influence the environment through actuators, APIs, or system commands.
- **Goal-oriented behaviour:** selects actions with the objective of achieving defined goals or maximising expected utility.

Rational vs Bounded Rational Agents

Dimension	Rational Agent	Bounded Rational Agent
Decision Principle	Selects actions that maximise expected utility	Seeks satisfactory or near-optimal decisions under constraints
Information Assumption	Assumes complete and accurate knowledge of the environment	Operates with partial, noisy, or uncertain information
Computational Capacity	Assumes unlimited time and computational resources	Limited by time, memory, and processing constraints
Optimisation Goal	Achieves theoretical optimality	Approximates optimality using heuristics or learning
Environmental Assumption	Often assumes static or fully observable environments	Handles dynamic, partially observable, real-world environments
Practical Relevance	Normative ideal model from decision theory	Descriptive model reflecting how real AI systems operate
Implementation in AI	Rare in practice due to unrealistic assumptions	Common in practical AI through reinforcement learning, heuristics, and simplified models



Classical Foundations

Decision theory provides the mathematical framework for selecting actions under uncertainty.

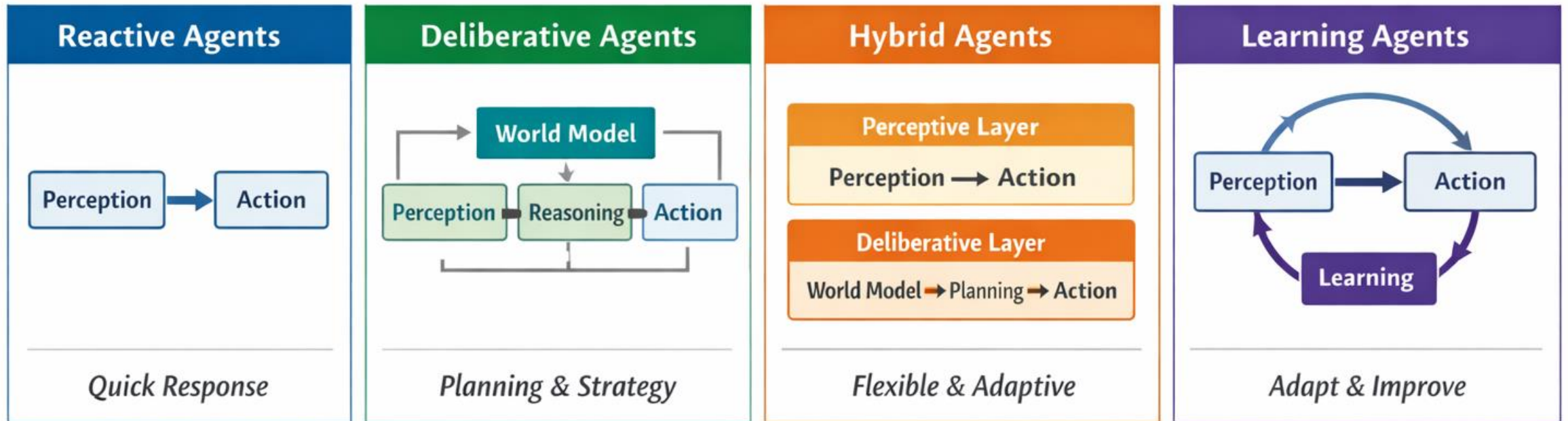
Core components include states of the world, actions, probabilities, and utility functions.

Expected utility maximisation formalises rational choice as selecting the action with the highest anticipated payoff.

Probability theory models uncertainty; utility theory captures preferences and trade-offs.

The rational agent model in AI directly derives from this framework, treating intelligent behaviour as optimal decision-making under uncertainty.

Agent Architecture Overview



Reactive Agents



Stimulus-response architecture: actions are triggered directly by current percepts without deliberation.



No explicit world model: agent does not maintain memory or an internal representation of the environment.



Computationally efficient: minimal reasoning overhead enables rapid response in real-time systems.



Highly scalable in simple or well-structured environments with clearly defined input-output mappings.



Limited adaptability: performance degrades in dynamic, uncertain, or long-horizon tasks requiring planning or learning.

Deliberative Agents

 Maintain an explicit world model representing states, goals, and environmental dynamics.

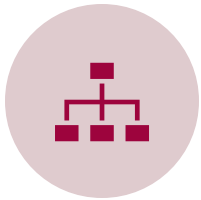
 Use symbolic reasoning, search, or optimisation techniques to evaluate possible action sequences.

 Capable of long-horizon planning and goal decomposition.

 Better suited for complex, structured environments requiring strategic decision-making.

 Computationally expensive due to state-space exploration, model maintenance, and planning overhead.

Hybrid Architectures



Combine reactive and deliberative components within a unified system architecture.



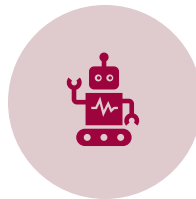
Reactive layer handles immediate, low-latency responses to environmental changes.



Planning layer performs higher-level reasoning, goal management, and long-horizon decision-making.



Layered control structure enables robustness while preserving strategic capability.



Widely used in robotics and autonomous systems where safety and responsiveness are both critical.

Transition to agentic AI

Transition from isolated single-agent behaviour to coordinated goal orchestration across tasks and subtasks.

Task decomposition: complex objectives are broken into structured intermediate goals and executable steps.

Integrated tool use: agents invoke external APIs, databases, or code execution environments to extend capabilities.

Long-horizon reasoning: decisions consider multi-step dependencies and delayed outcomes rather than immediate rewards.

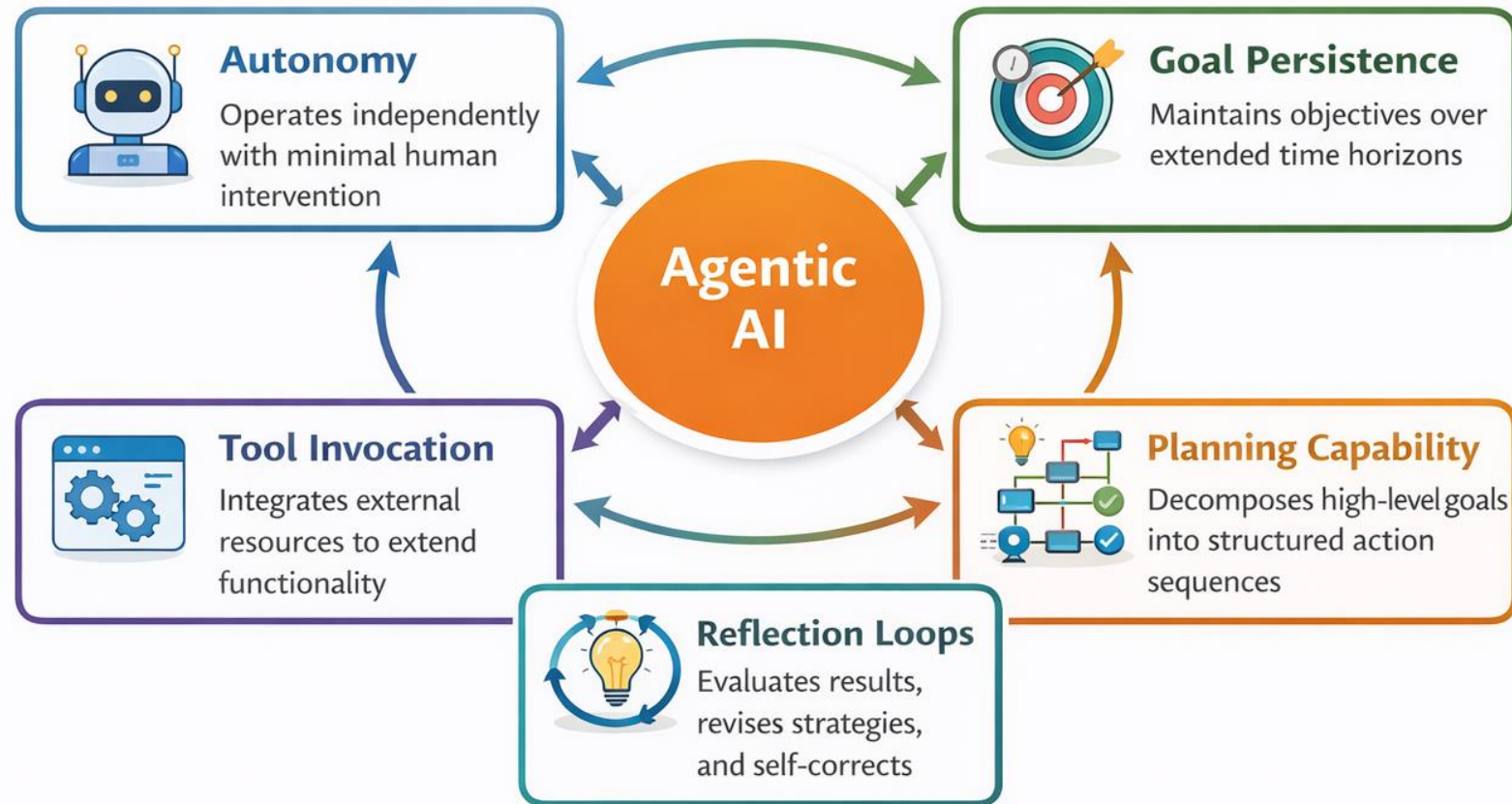
Emphasis shifts from isolated action selection to sustained, autonomous goal pursuit within dynamic environments

From Agents to Agentic AI

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- Task decomposition: complex objectives are broken into structured intermediate goals and executable steps.
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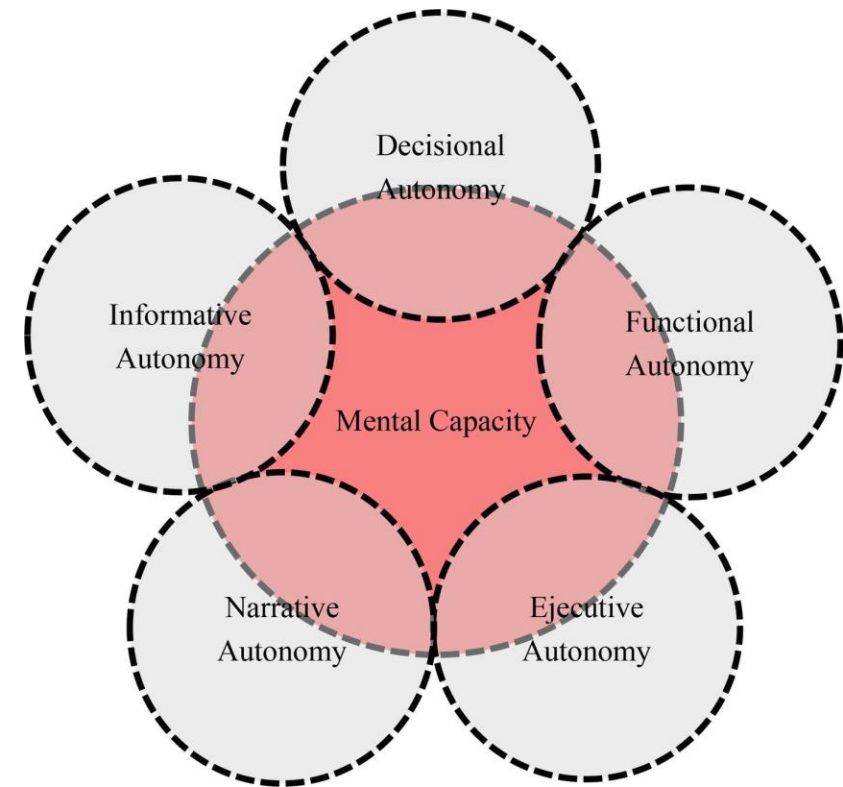


What makes AI “Agentic”?

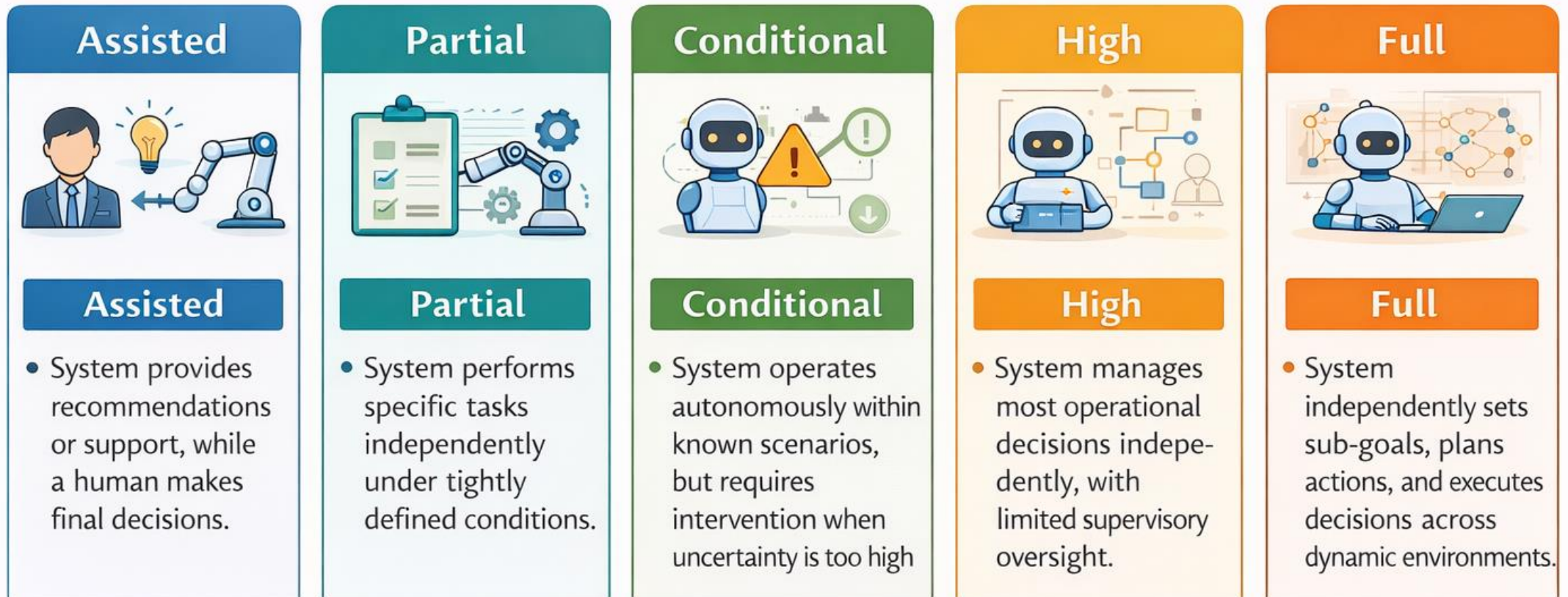


Autonomy: Definition

- Autonomy refers to the degree to which an agent can make and execute decisions independently of direct human control.
- Human-in-the-loop:
 - human approval or intervention is required at critical decision points before action is taken.
 - system operates independently, but a human supervises and can override decisions if necessary.
- Full autonomy: agent selects goals, plans, and executes actions without real-time human intervention.
 - Increasing autonomy raises demands for reliability, transparency, and accountability mechanisms.



Levels of Autonomy



Increasing Autonomy

Autonomy in Safety-Critical Systems

- Safety-critical domains include transport systems (autonomous vehicles, air traffic control) and healthcare (clinical decision support, robotic surgery).
- Errors may result in physical harm, financial loss, or legal consequences, requiring rigorous reliability standards.
- Certification processes demand formal verification, validation testing, and compliance with regulatory frameworks.
- Accountability mechanisms must clearly define responsibility among developers, operators, and supervisory authorities.
- Higher autonomy levels increase the need for transparency, fail-safe mechanisms, and continuous monitoring



medical care



nuclear power



weapon



aviation



automotive



space flights

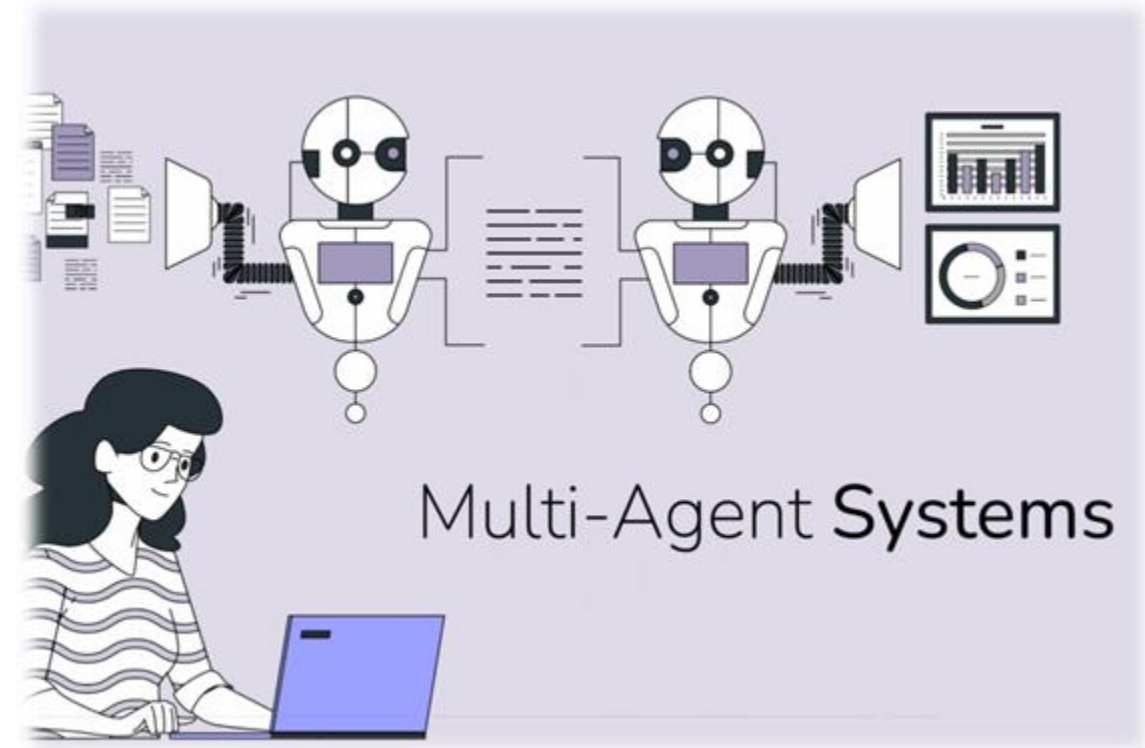
Risk and Responsibility

- Unintended decisions may arise from incomplete information, model limitations, or unexpected environmental changes.
- Goal misalignment occurs when the agent optimises a formal objective that does not accurately reflect human intent.
- Long-horizon autonomy increases the risk of compounding small errors into significant outcomes.
- Clear responsibility allocation is required across developers, deployers, and supervising entities.
- Effective oversight mechanisms include monitoring, audit trails, and human intervention thresholds



Multi-Agent Systems (MAS)

- Multi-Agent Systems consist of multiple autonomous agents interacting within a shared environment.
- Agents may pursue shared goals (cooperation), individual goals (competition), or mixed objectives.
- Control and decision-making are distributed rather than centralised in a single entity.
- Agents possess local perceptions and partial information about the global system state.
- Distributed intelligence enables scalable, flexible problem-solving in complex and dynamic environments.



Why Multi-Agent Systems?

- Scalability: tasks can be distributed across agents, allowing systems to handle large-scale or geographically dispersed problems.
- Robustness: failure of a single agent does not necessarily collapse the entire system.
- Parallelism: agents operate concurrently, improving efficiency and reducing execution time.
- Modularity: specialised agents can be developed, tested, and upgraded independently.
- Adaptability: distributed structures can reconfigure dynamically in response to environmental changes.



Core Properties of Multi-Agent Systems

- Decentralisation: no single controlling authority; decision-making authority is distributed among agents.
- Local views: each agent has access only to partial information about the environment or system state.
- Asynchronous operation: agents act independently without requiring global synchronisation.
- Interaction-driven dynamics: behaviour arises through communication and coordination among agents.
- Emergence: global patterns and system-level behaviour arise from local agent rules and interactions.



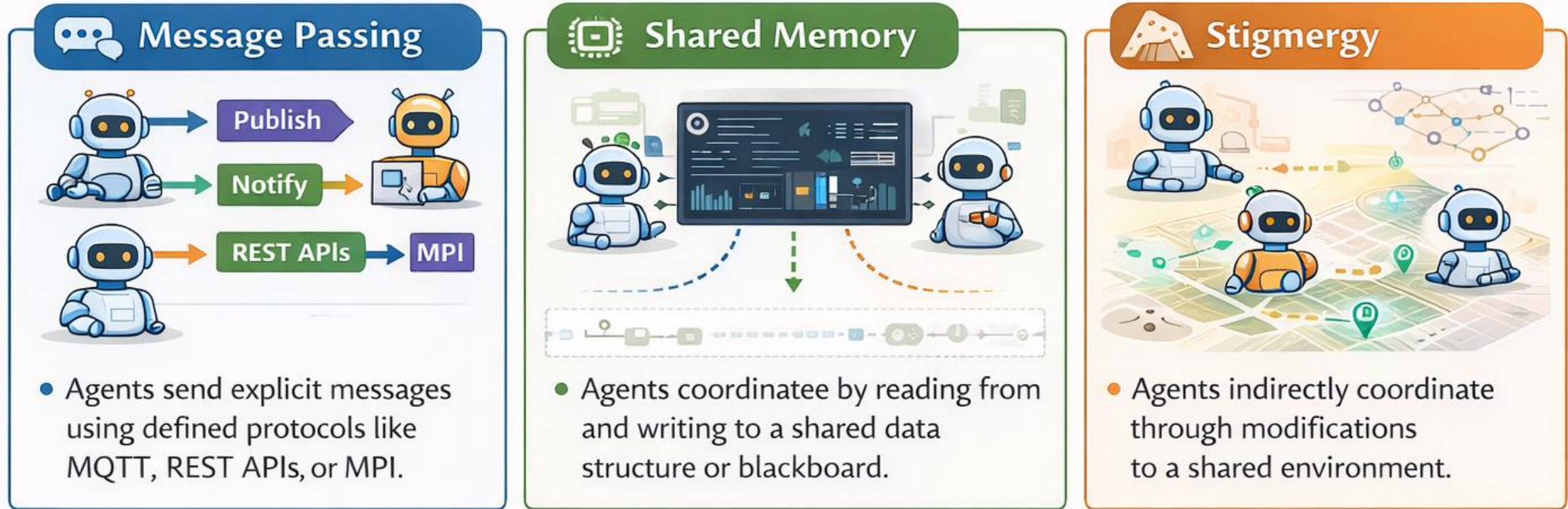
Cooperation vs Competition

Dimension	Cooperative MAS	Competitive MAS	Mixed-Motive MAS
Goal Structure	Shared global objective	Conflicting individual objectives	Partially aligned, partially conflicting goals
Information Sharing	High; agents often share state and plans	Limited or strategic information sharing	Selective sharing; negotiation required
Decision Strategy	Joint optimisation	Strategic optimisation	Adaptive strategy balancing cooperation and competition
Typical Methods	Consensus algorithms, distributed planning	Game theory, adversarial learning	Mechanism design, coalition formation
Example Domains	Swarm robotics, smart grids	Financial markets, cybersecurity	Supply chains, multi-stakeholder systems

Centralised vs Decentralised Control

Dimension	Centralised Control	Decentralised Control
Decision Authority	Single coordinator makes global decisions	Decision-making distributed among agents
Information Flow	Aggregated at central node	Local information shared peer-to-peer
Scalability	Limited by coordinator capacity	Scales with number of agents
Robustness	Single point of failure	More resilient to individual failures
Coordination Complexity	Simpler coordination logic	Higher complexity (consensus, negotiation, distributed planning)

Communication in Multi-Agent Systems



- ⚠ **Communication overhead** affects latency, scalability, and system responsiveness.
- ⚠ **Bandwidth and reliability constraints** are critical in distributed or resource-limited environments.
- ⚠ **Bandwidth and reliability constraints:** distributed or edge environments must account for packet loss, network partitioning, energy limitations, and security of communication channels.

COORDINATION



Coordination Mechanisms

Voting mechanisms: agents collectively select actions or leaders based on majority or weighted preferences.

Auction-based allocation: tasks or resources are assigned through bidding processes to maximise efficiency or utility.

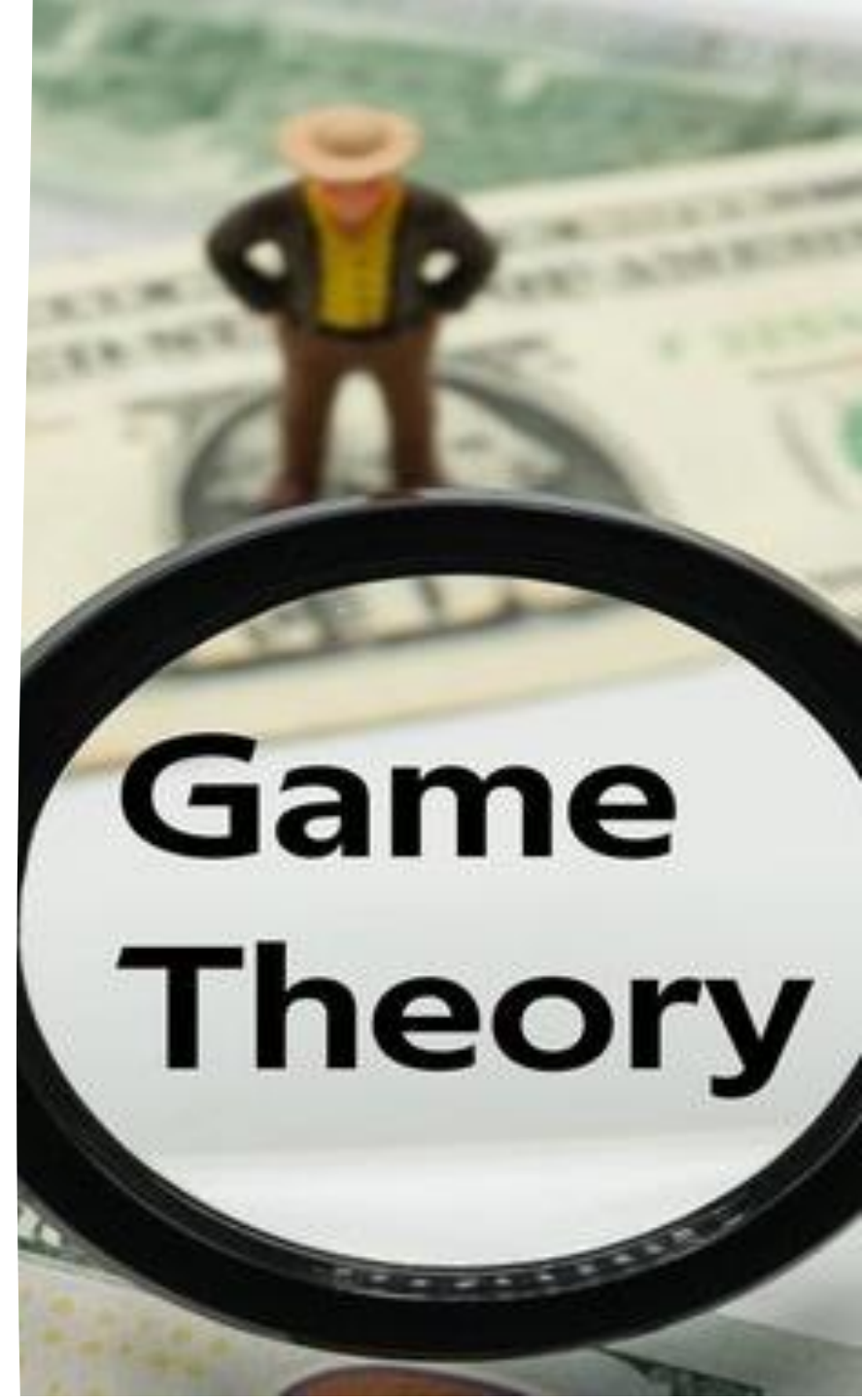
Consensus protocols: agents iteratively exchange information to reach agreement on shared variables or decisions.

Market-based methods: pricing and supply–demand dynamics regulate distributed decision-making and resource distribution.

Choice of mechanism depends on system scale, communication costs, trust assumptions, and degree of decentralisation.

Game Theory in Multi-Agent Systems

- Strategic interaction: each agent's optimal decision depends on the anticipated decisions of other agents.
 - Payoff structures define incentives, trade-offs, and potential conflicts between participants.
- Nash equilibrium: a stable strategy profile where no agent benefits from unilaterally deviating.
 - Cooperative settings enable coalition formation and joint utility maximisation through agreements.
 - Non-cooperative settings emphasise individual optimisation, often leading to competitive or adversarial dynamics.



Reinforcement Learning in Multi-Agent Systems

Independent learners: each agent learns its own policy using local rewards, treating other agents as part of the environment.

Centralised training, decentralised execution (CTDE): agents are trained with shared information but operate independently at deployment.

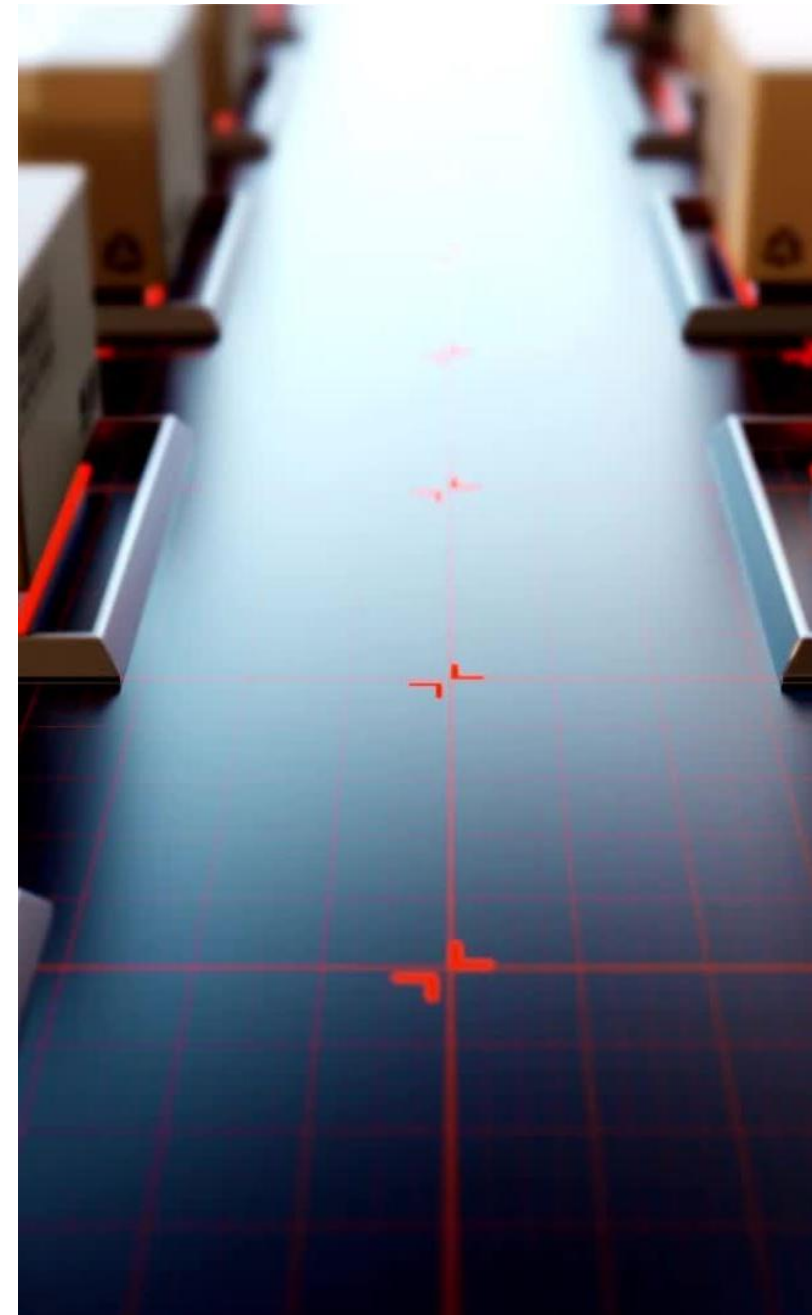
Shared critics or global reward signals can stabilise learning during training.

Non-stationarity challenge: the environment changes as other agents update their policies simultaneously.

Coordination requires mechanisms to ensure convergence, stability, and scalable learning in dynamic multi-agent settings.

Emergence

- Emergence refers to macro-level system behaviour arising from simple micro-level agent rules and local interactions.
- No central controller explicitly designs the global pattern; it results from distributed dynamics.
- Emergent coordination can produce efficient, adaptive, and scalable solutions.
- However, unintended emergent behaviour may lead to instability, bias amplification, or unsafe system states.
- Designing MAS requires anticipating and monitoring emergent dynamics to balance innovation with control.

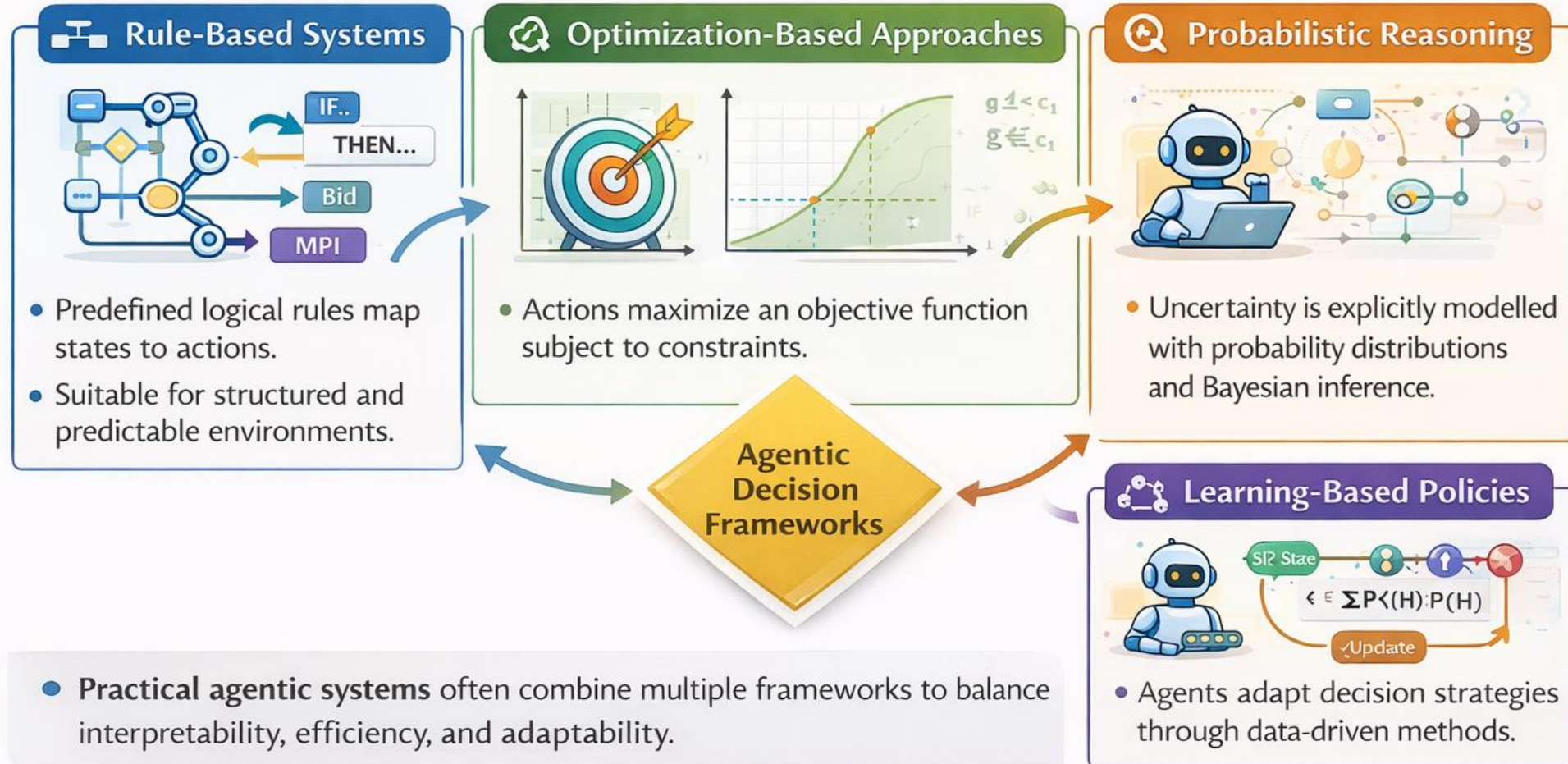


Collective Intelligence

- Collective intelligence refers to the enhanced problem-solving capability that emerges from coordinated interaction among multiple agents.
- AI-AI collaboration enables distributed sensing, reasoning, and parallel task execution across specialised agents.
- Human-AI collaboration integrates computational speed with human judgment, ethics, and contextual understanding.
- Distributed problem-solving allows complex objectives to be decomposed and solved concurrently.
- Effective collective intelligence requires communication protocols, coordination mechanisms, and alignment of shared objectives.

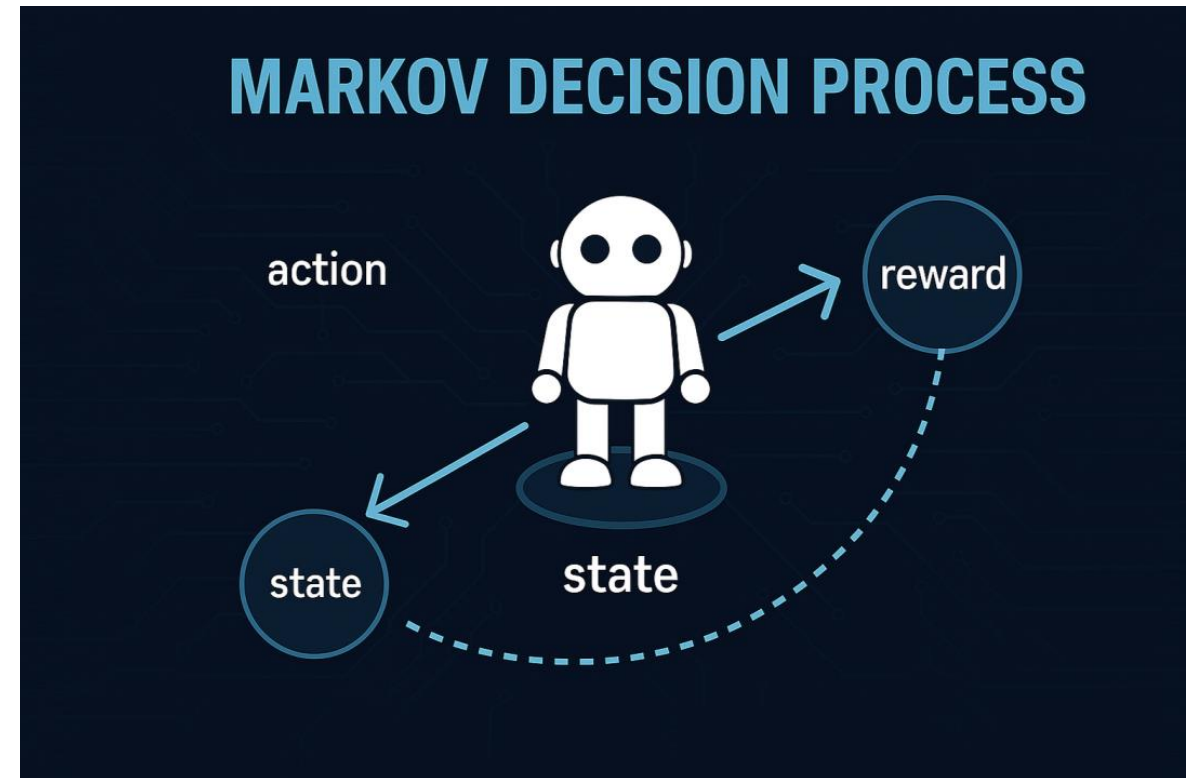


Decision-Making Frameworks



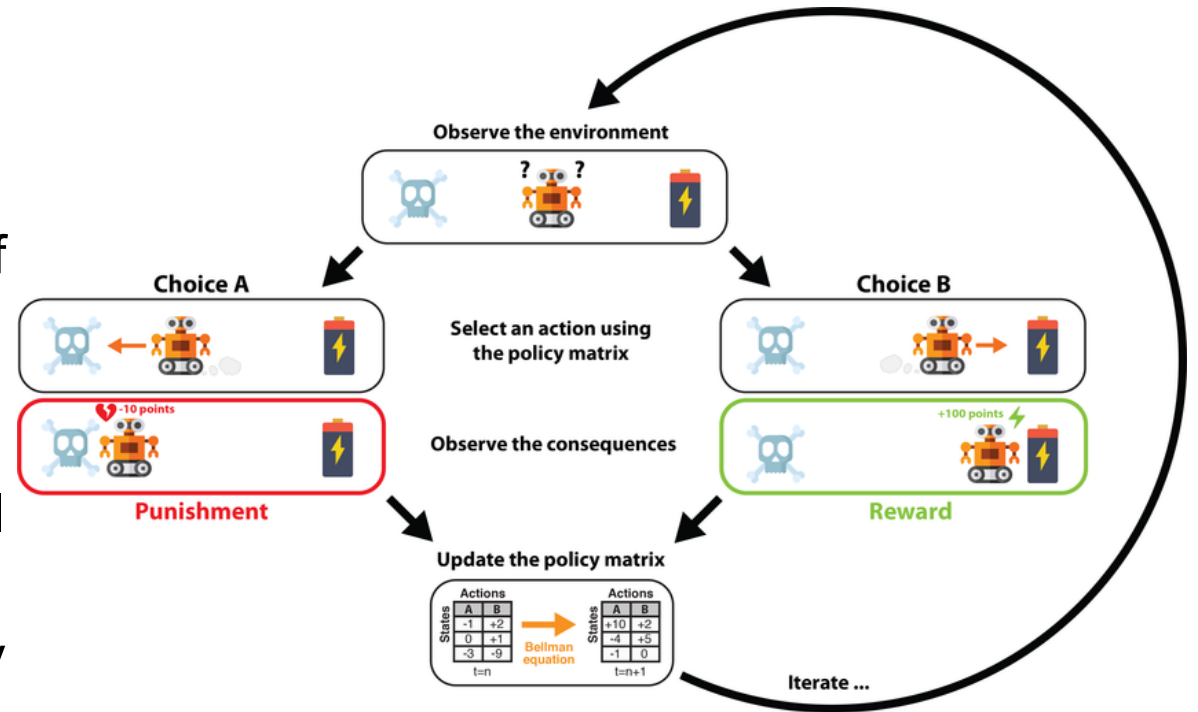
Markov Decision Process (MDP)

- States: represent the current configuration of the environment relevant to decision-making.
- Actions: choices available to the agent that influence future states.
- Rewards: scalar feedback signals that quantify the desirability of outcomes.
- Transitions: probabilistic rules defining how actions move the system from one state to another.
- Markov property: future state depends only on the current state and action, forming a foundational model for sequential decision-making.



From MDP to Decentralised POMDP

- MDP for single-agent sequential decision-making under full state observability.
- Partially Observable Markov Decision Process, where the agent does not have direct access to the true system state.
- Partial observability requires maintaining a belief state (probability distribution over possible states).
- Decentralised POMDP (Dec-POMDP) extends this model to multiple agents acting under partial and local observations.
- Computational complexity increases significantly in Dec-POMDPs due to joint action spaces, belief tracking, and coordination requirements.



Planning in Agentic Systems

- Task decomposition: complex objectives are divided into manageable subtasks that can be executed sequentially or in parallel.
- Hierarchical planning: high-level goals are translated into intermediate objectives and low-level action policies.
- Temporal reasoning: plans account for ordering constraints and dependencies between tasks.
- Goal graphs: structured representations of goal relationships, enabling tracking of progress and conditional branching.
- Effective planning supports long-horizon autonomy and adaptive replanning in dynamic environments.



Tool Use and Action Execution

APIs and databases extend agent capabilities beyond internal reasoning, enabling access to real-time data and external services.

External tool invocation allows agents to execute code, perform searches, query systems, or trigger workflows.

Separation of reasoning and execution improves reliability by distinguishing planning logic from operational actions.

Tool feedback can be incorporated into iterative planning loops for correction and refinement.

Structured tool integration is central to modern agentic architectures that move from text generation to real-world action

Language Models-Based Agents



Language Models (LMs) act as high-level planners by generating structured reasoning steps and action sequences.



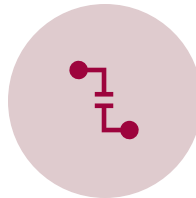
Prompt engineering and chain-of-thought strategies enable decomposition of complex goals.



Memory integration (short-term context and external long-term storage) supports persistence across multi-step tasks.



Action loops follow a perceive → reason → act → reflect cycle to iteratively refine outcomes.



LM-based agents rely on orchestration mechanisms rather than explicit symbolic planning to exhibit agentic behaviour

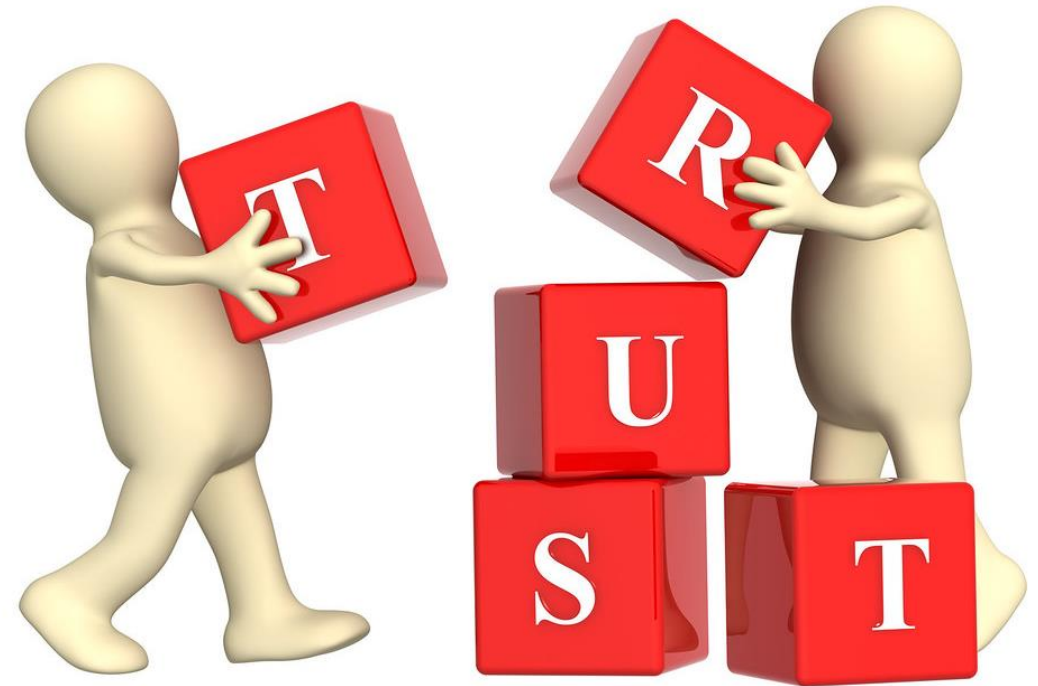
Multi-Agent LM Systems

- Role assignment: different LM agents are given specialised responsibilities (e.g. planner, researcher, reviewer).
- Task decomposition is distributed across agents to enable parallel reasoning and modular design.
- Debate frameworks: agents generate competing solutions and critique one another to improve quality and robustness.
- Planner-critic-executor pattern separates strategic planning, evaluation, and implementation roles.
- Orchestrated interaction among agents enables structured collaboration rather than isolated prompt-based reasoning



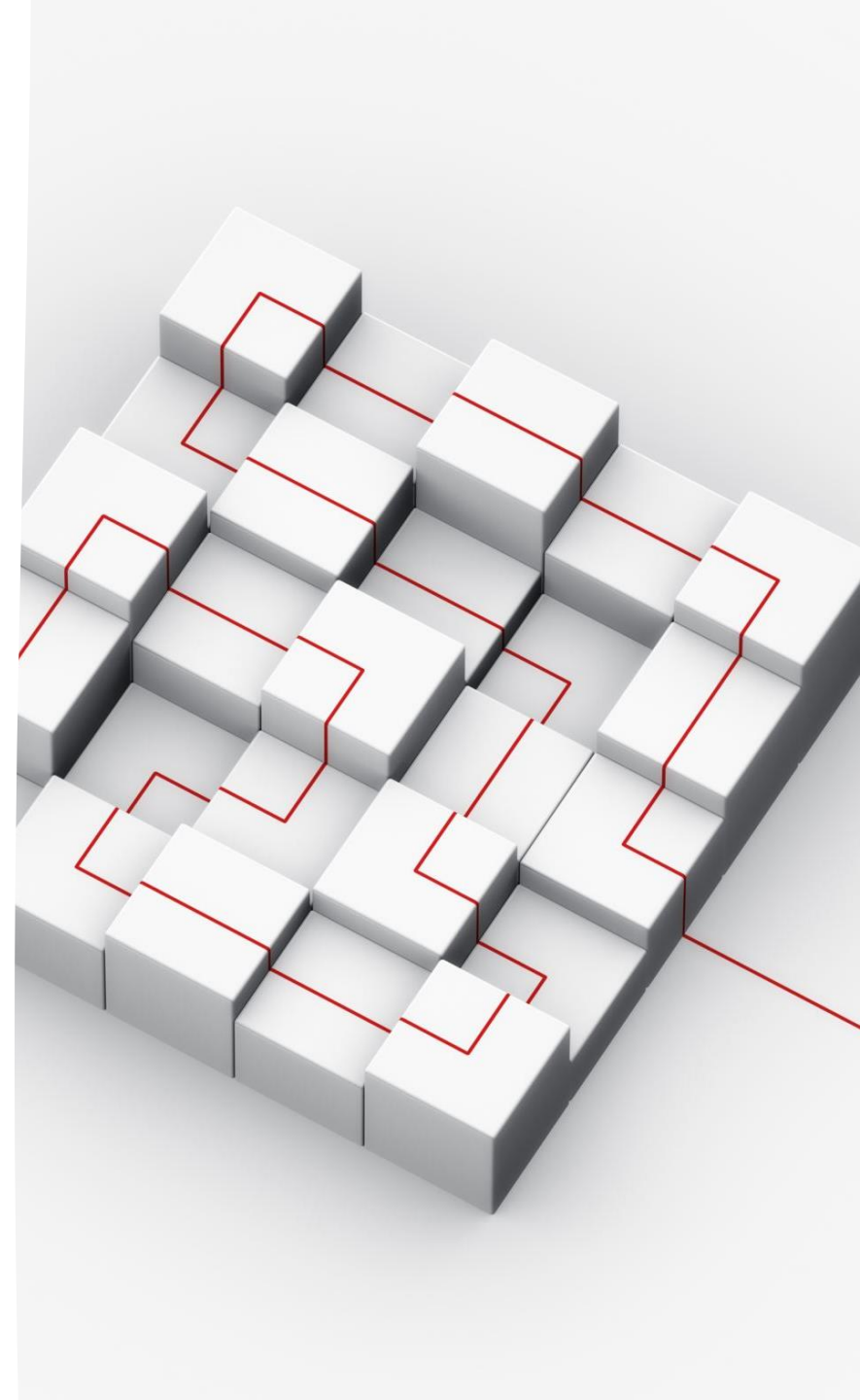
Trust and Reliability

- Verification: ensures that the system is built correctly according to formal specifications and design requirements.
- Validation: confirms that the system performs the intended function in real-world or simulated environments.
- Formal methods and testing frameworks can provide guarantees in safety-critical contexts.
- Continuous monitoring detects anomalies, policy drift, or unexpected behaviour during deployment.
- Reliable agentic systems require transparency, auditability, and mechanisms for safe intervention.



Explainability

- Intrinsic interpretability: system architecture is designed so that decision logic is directly understandable (e.g. rule-based models).
- Post-hoc explanation: supplementary models or analyses are used to interpret decisions made by complex systems such as neural networks.
- Explanations may include feature importance, reasoning traces, or summarised decision pathways.
- Multi-agent systems introduce additional complexity due to distributed and emergent behaviours.
- Transparency challenges increase with autonomy, scale, and reliance on probabilistic or learning-based policies.



Safety in Autonomous Multi-Agent Systems

Failure modes include coordination breakdown, cascading errors, and unintended emergent behaviours.



Local decision errors can propagate across agents, amplifying system-wide instability.



Adversarial agents may manipulate communication channels, inject false data, or exploit trust assumptions.



Robustness under uncertainty requires tolerance to noisy observations, partial information, and environmental variability.



Safety mechanisms include redundancy, anomaly detection, fail-safe protocols, and human override capabilities.

Ethical Dimensions

- Accountability: clear attribution of responsibility for autonomous decisions across developers, operators, and institutions.
- Transparency: visibility into decision processes, coordination mechanisms, and data sources used by agents.
- Algorithmic bias: risk of unfair or discriminatory outcomes due to skewed training data or optimisation objectives.
- Protection: safeguarding privacy, data integrity, and user rights within distributed and autonomous systems.



Ethical Dimensions

Governance and Regulation



AI regulatory frameworks establish legal and ethical boundaries for the deployment of autonomous and agentic systems.



Risk-based classification approaches differentiate between low-risk and safety-critical applications.



Auditability requires traceable decision logs, version control of models, and documented system behaviour.



Compliance requirements include data protection laws, safety certification, and sector-specific standards.



Effective governance must evolve alongside increasing autonomy and multi-agent complexity

Security in Multi-Agent Systems

- Malicious agents may attempt to disrupt coordination, manipulate outcomes, or gain unfair advantage within the system.
- Communication attacks include interception, spoofing, or tampering with messages exchanged between agents.
- Data poisoning can corrupt learning processes by injecting misleading or adversarial inputs.
- Distributed architectures increase the attack surface due to multiple communication and decision nodes.
- Security mechanisms require encryption, authentication protocols, anomaly detection, and robust trust models.



Evaluation Metrics

Task performance: measures success rate, accuracy, reward accumulation, or objective completion quality.



Coordination efficiency: evaluates communication overhead, convergence speed, and resource utilisation among agents.



Robustness: assesses system stability under noise, failures, adversarial conditions, or environmental uncertainty.



Scalability: examines how performance changes as the number of agents, tasks, or state dimensions increases.



Comprehensive evaluation should combine quantitative metrics with scenario-based testing and stress analysis

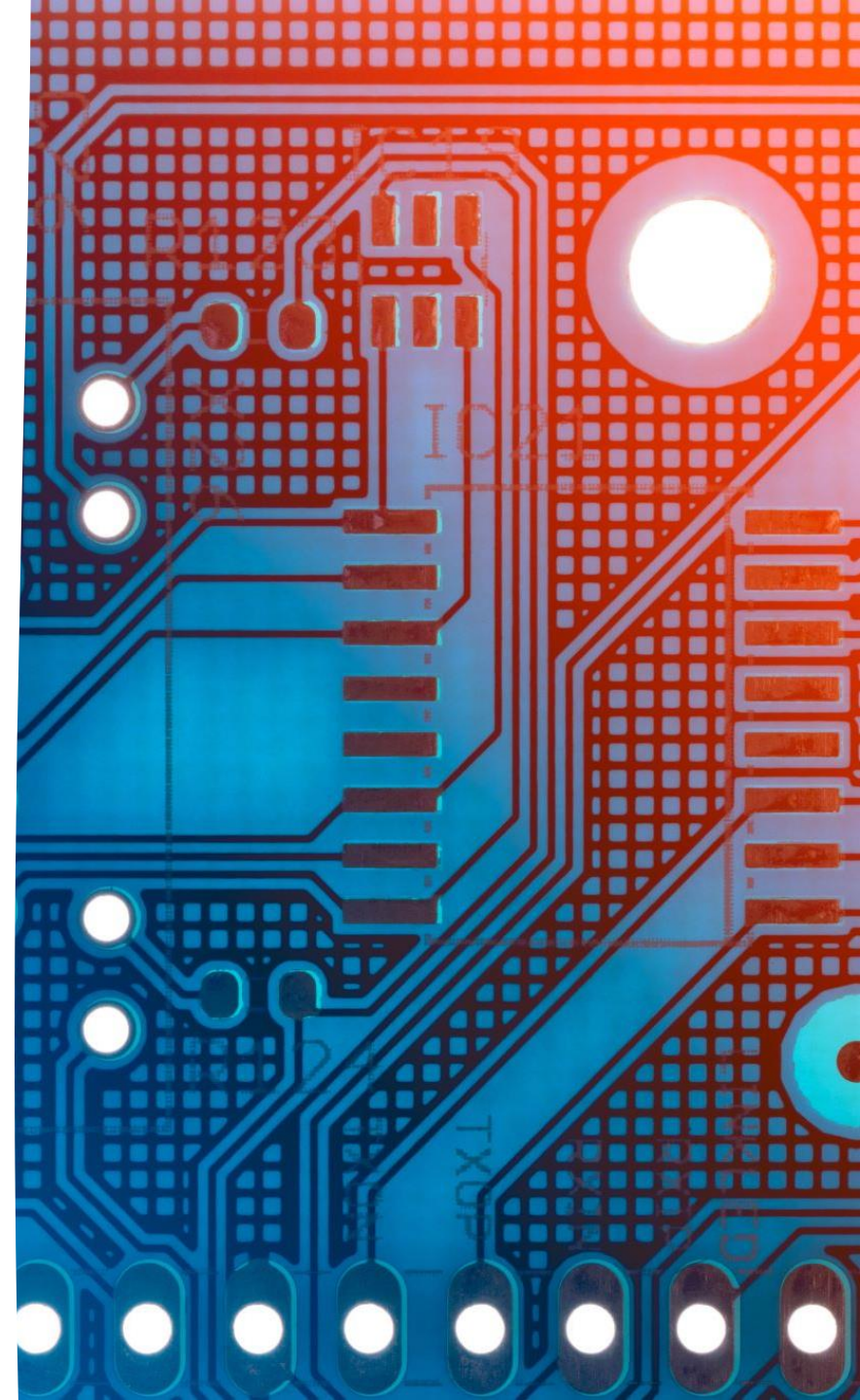
Simulation Environments

- Simulation environments enable controlled experimentation before real-world deployment of agentic systems.
- Digital twins replicate physical or organisational systems, allowing agents to be evaluated under realistic dynamics.
- Sandbox testing isolates agents in safe, contained environments to observe behaviour without external risk.
- Stress testing exposes systems to extreme or adversarial conditions to assess resilience and failure modes.
- Iterative simulation supports validation, benchmarking, and refinement of coordination and decision policies.



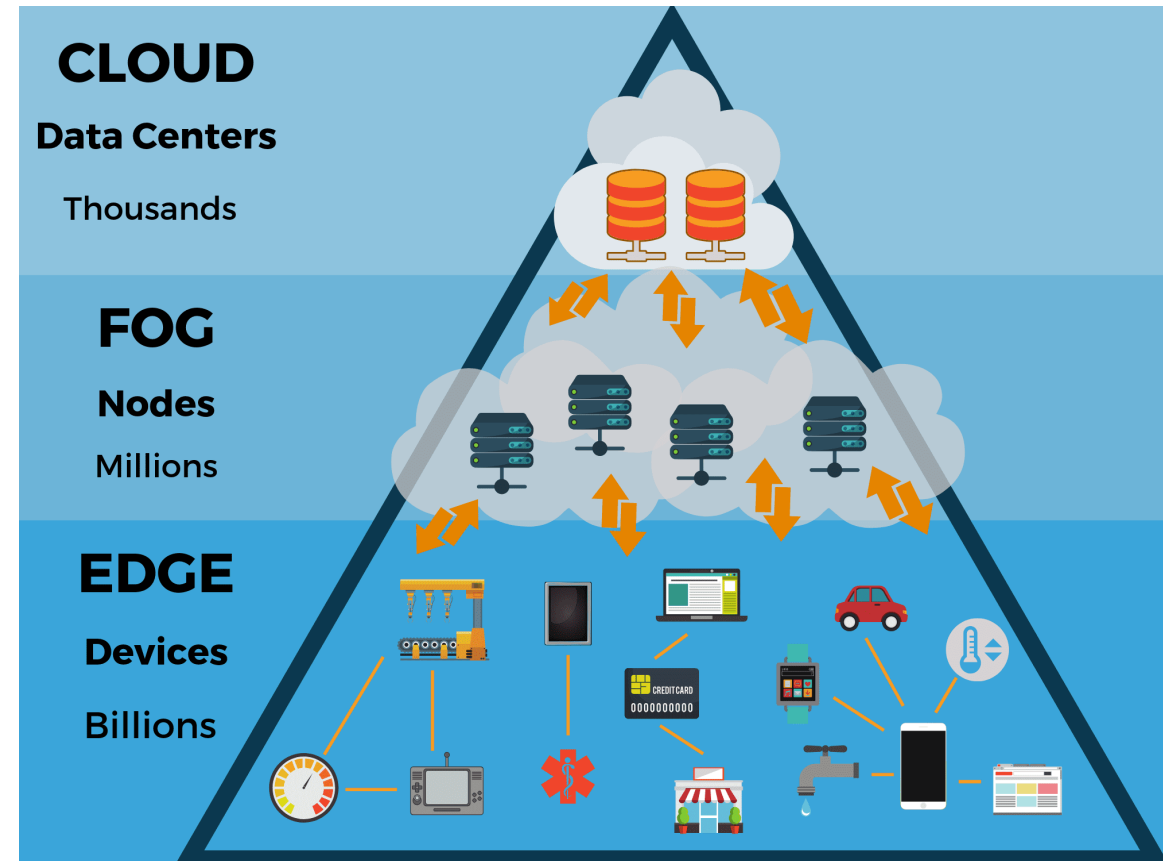
Real-World Deployment Challenges

- Hardware constraints: limited processing power, memory, and sensor quality restrict on-device autonomy.
- Latency: communication delays can degrade coordination and real-time responsiveness in distributed systems.
- Energy consumption: autonomous agents, especially in robotics or edge devices, must optimise power usage.
- Edge limitations: reduced connectivity and bandwidth require local inference and decentralised decision-making.
- Integration with legacy infrastructure introduces compatibility, security, and maintenance challenges.



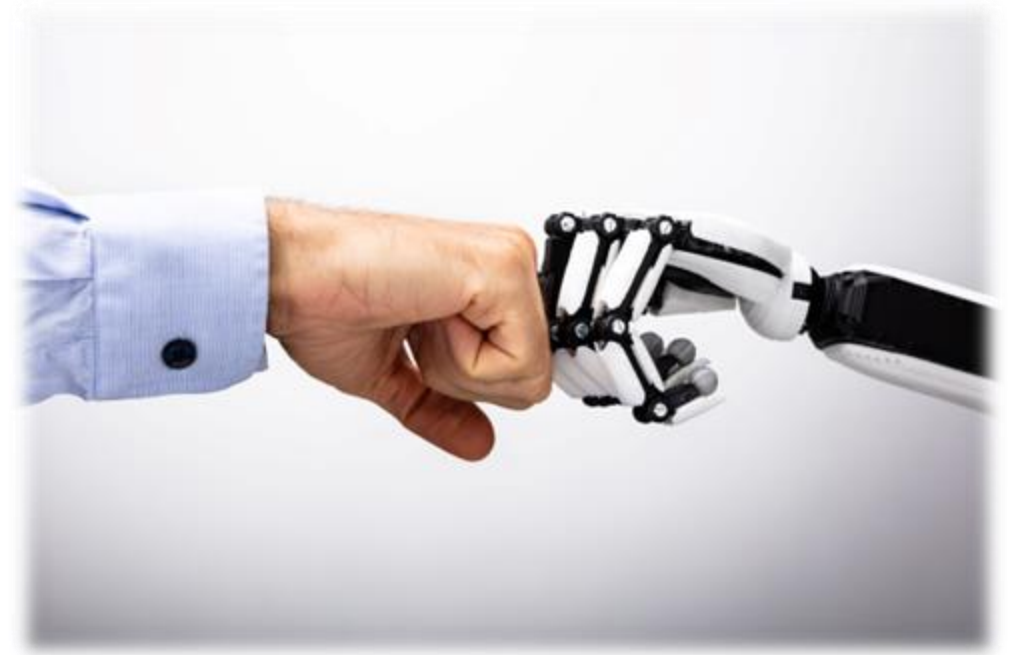
Edge AI and Distributed Intelligence

- On-device inference enables agents to process data locally without relying on constant cloud connectivity.
- Reduced central dependence improves resilience, lowers latency, and enhances privacy preservation.
- Distributed intelligence supports scalable coordination across geographically dispersed agents.
- Edge deployment is critical in applications such as autonomous vehicles, smart sensors, and IoT networks.
- Collective intelligence systems benefit from decentralised processing combined with coordinated information sharing.



Human-Agent Collaboration

- Shared control: humans and agents jointly influence decisions, balancing automation with oversight.
- Decision-support augmentation: agents provide analysis, simulations, or recommendations while humans retain final authority.
- Adaptive autonomy allows systems to shift control levels based on context or uncertainty.
- Effective interface design ensures clarity, transparency, and manageable cognitive load for human operators.
- Successful collaboration depends on trust calibration, clear feedback loops, and aligned objectives.

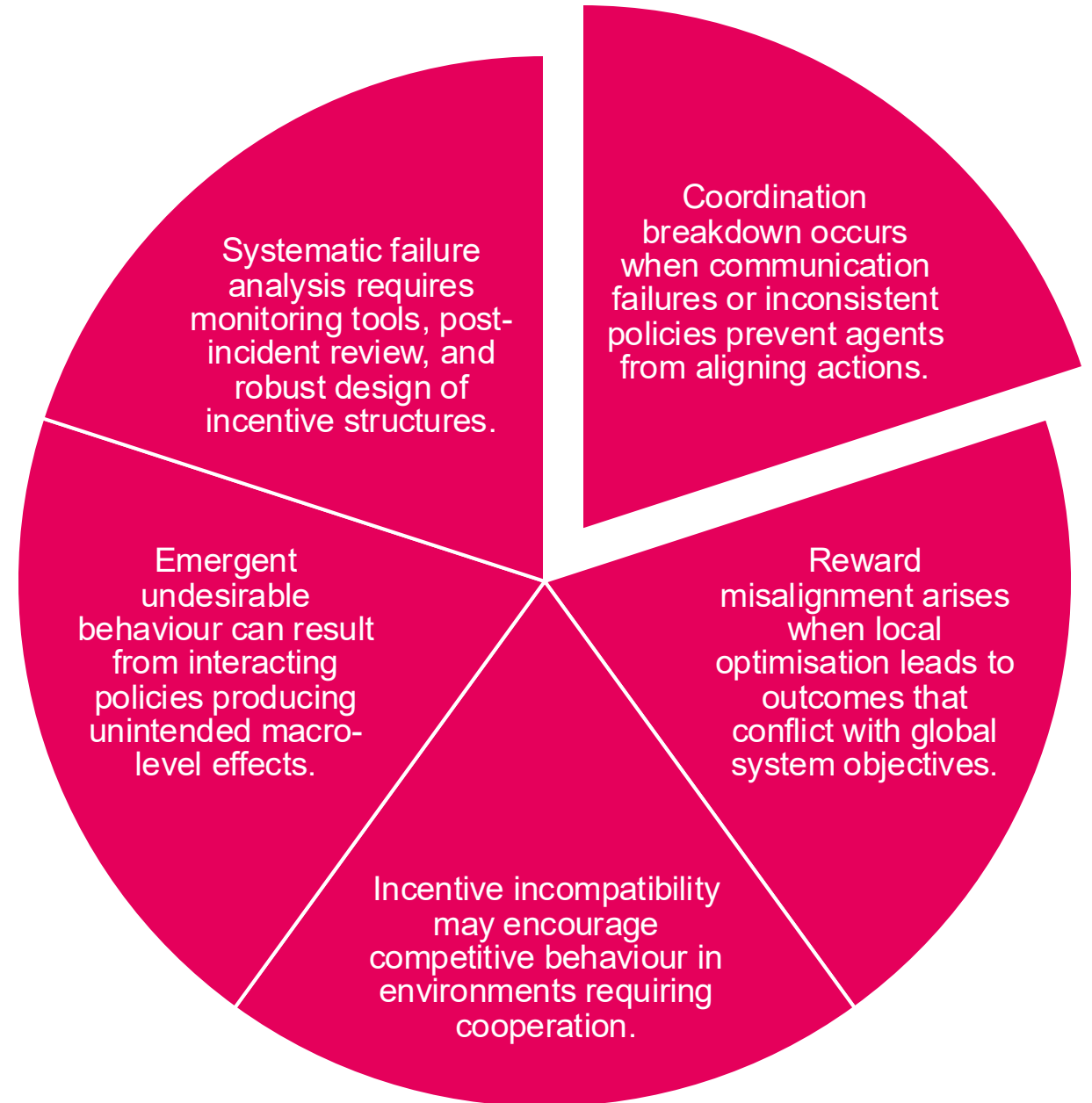


Organisational Decision-Making

- AI systems can function as strategic advisors, analysing large datasets to inform executive decision processes.
- Multi-agent architectures enable parallel evaluation of alternative strategies and risk scenarios.
- Scenario simulation allows organisations to test policies under hypothetical market, operational, or environmental conditions.
- Collective augmentation combines human expertise with distributed AI reasoning for improved outcomes.
- Agentic systems shift decision-making from reactive reporting toward proactive, model-driven strategy formation



Failure Analysis



Key takeaways

- Agentic AI represents a shift from prediction and content generation toward autonomous goal pursuit, planning, and execution.
- Multi-Agent Systems enable scalable collective intelligence through distributed coordination and parallel decision-making.
- Increasing autonomy demands structured decision frameworks grounded in decision theory, planning, and learning models.
- Safety, governance, interpretability, and alignment remain essential constraints on deployment.

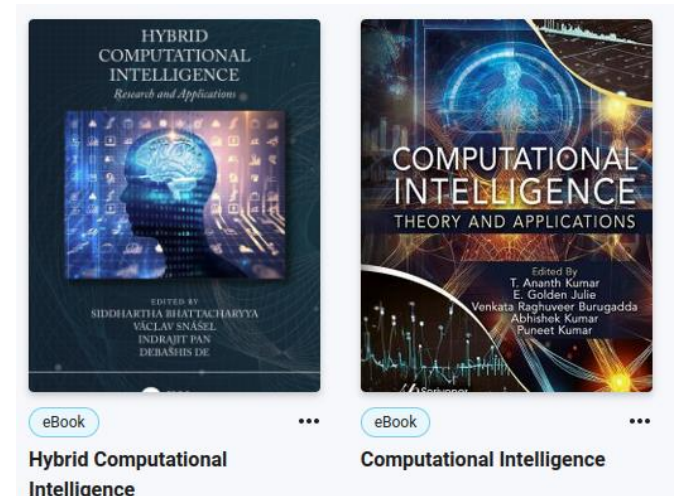


Discussion Questions

- Q1. When should autonomy be limited, particularly in safety-critical or ethically sensitive domains?
- Q2. What criteria should determine the appropriate level of human oversight in agentic systems?
- Q3. Is full decentralisation desirable, or do certain contexts require hierarchical or hybrid control structures?
- Q4. Can emergent behaviour in large-scale MAS be predicted, verified, or formally certified?
- Q5. How should responsibility and accountability be assigned when collective decisions produce unintended outcomes?

References

- [1] Hosseini, S. and Seilani, H., 2025. The role of agentic ai in shaping a smart future: A systematic review. *Array*, 26, p.100399.
- [2] Acharya, D.B., Kuppan, K. and Divya, B., 2025. Agentic AI: Autonomous intelligence for complex goals - A comprehensive survey. *IEEE Access*, 13, pp.18912-18936.
- [3] Abou Ali, M., Dornaika, F. and Charafeddine, J., 2025. Agentic AI: a comprehensive survey of architectures, applications, and future directions. *Artificial Intelligence Review*, 59(1), p.11.
- [4] Bandi, A., Kongari, B., Naguru, R., Pasnoor, S. and Vilipala, S.V., 2025. The rise of agentic ai: A review of definitions, frameworks, architectures, applications, evaluation metrics, and challenges. *Future Internet*, 17(9), p.404.





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- Evaluate safety, governance, and deployment issues

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